Cybersecurity



Capstone Project



Implementing and Managing Active Directory in Windows Server

Objective



To set up and manage Active Directory in Windows server, including configuring domain controllers, structuring organizational units, managing user accounts, and implementing security policies

Description:

This project highlights the value of centralized identity management, security, and resource allocation. Implementing Active Directory enhances authentication, enforces security policies, and simplifies user and group management for better cybersecurity and efficiency.

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Problem Statement and Motivation



Real-time scenario:

Organizations often struggle with managing user authentication, resource access, and security policies in large-scale IT environments. Without a structured Active Directory (AD) implementation, companies face challenges such as unauthorized access, inefficient user management, and weak policy enforcement, leading to security vulnerabilities and operational inefficiencies.

This project focuses on deploying and managing Active Directory in a Windows server environment to centralize authentication, define organizational units, and enforce security policies through Group Policy Objects (GPOs).

By structuring the AD environment, the project enhances network security, streamlines administrative tasks, and ensures compliance with enterprise security standards. The goal is to improve access control, reduce security risks, and optimize IT management in an enterprise setting.

Industry Relevance



The following tools used in this project serve specific purposes within the industry:

1. **Windows Servers:** A widely used OS for managing Active Directory and enterprise networks

2. **Active Directory Domain Services (AD DS)**: Enables centralized user authentication and policy enforcement

3. **Group Policy Objects (GPOs):** Used to enforce security policies and system configurations

4. **Windows Defender Firewall**: Protects networks by enforcing security rules and monitoring traffic

Tasks

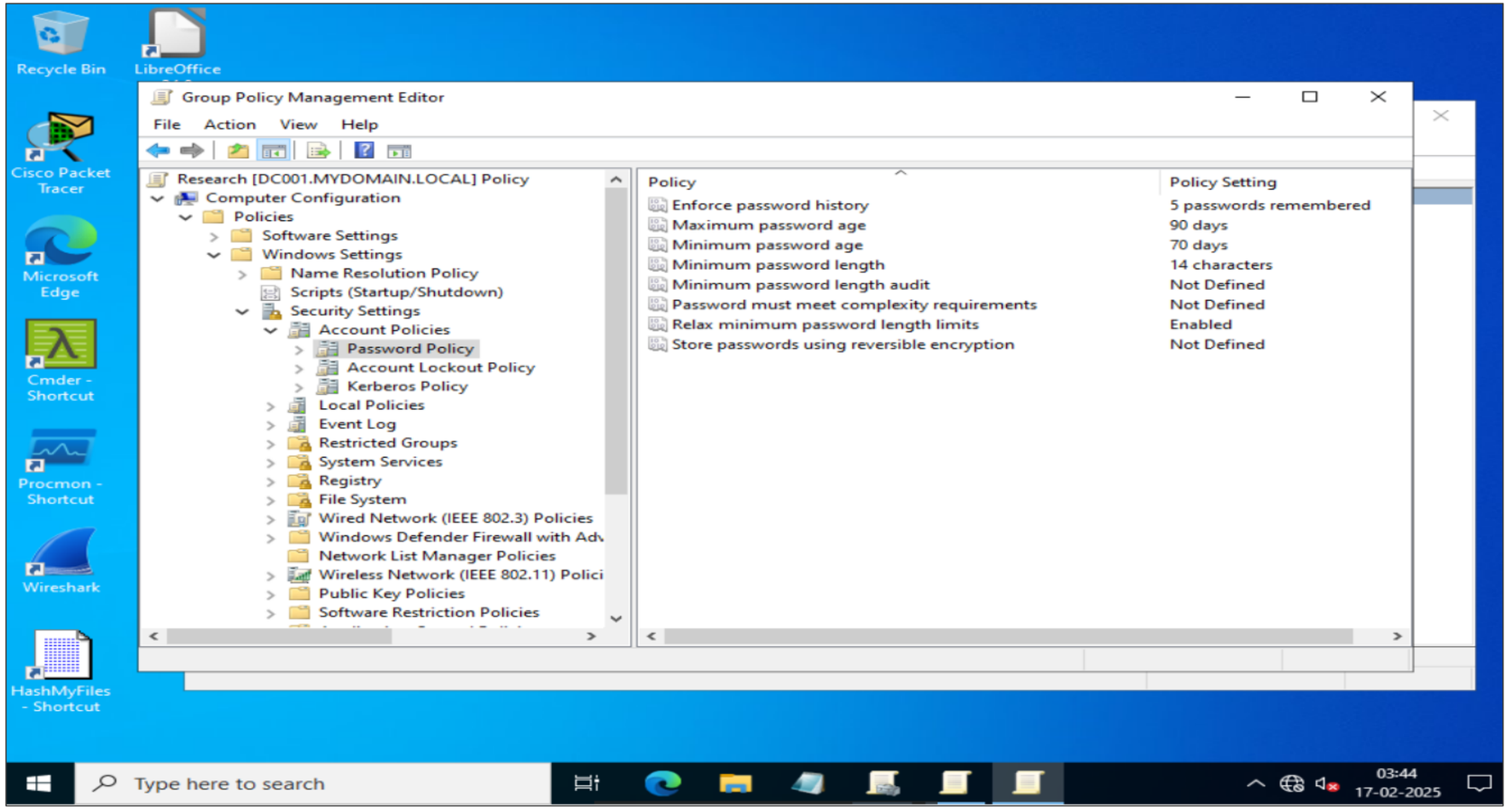


The following tasks outline the process of performing vulnerability exploitation and applying remediation strategies:

1. Install and configure Active Directory Domain Services (AD DS) to establish a domain controller
2. Create Organizational Units (OUs) and define user roles and security groups
3. Implement Group Policy Objects (GPOs) to enforce security and access policies
4. Configure client systems to join the domain and authenticate using AD credentials
5. Apply security measures, including password policies, login restrictions, and account lockout policies

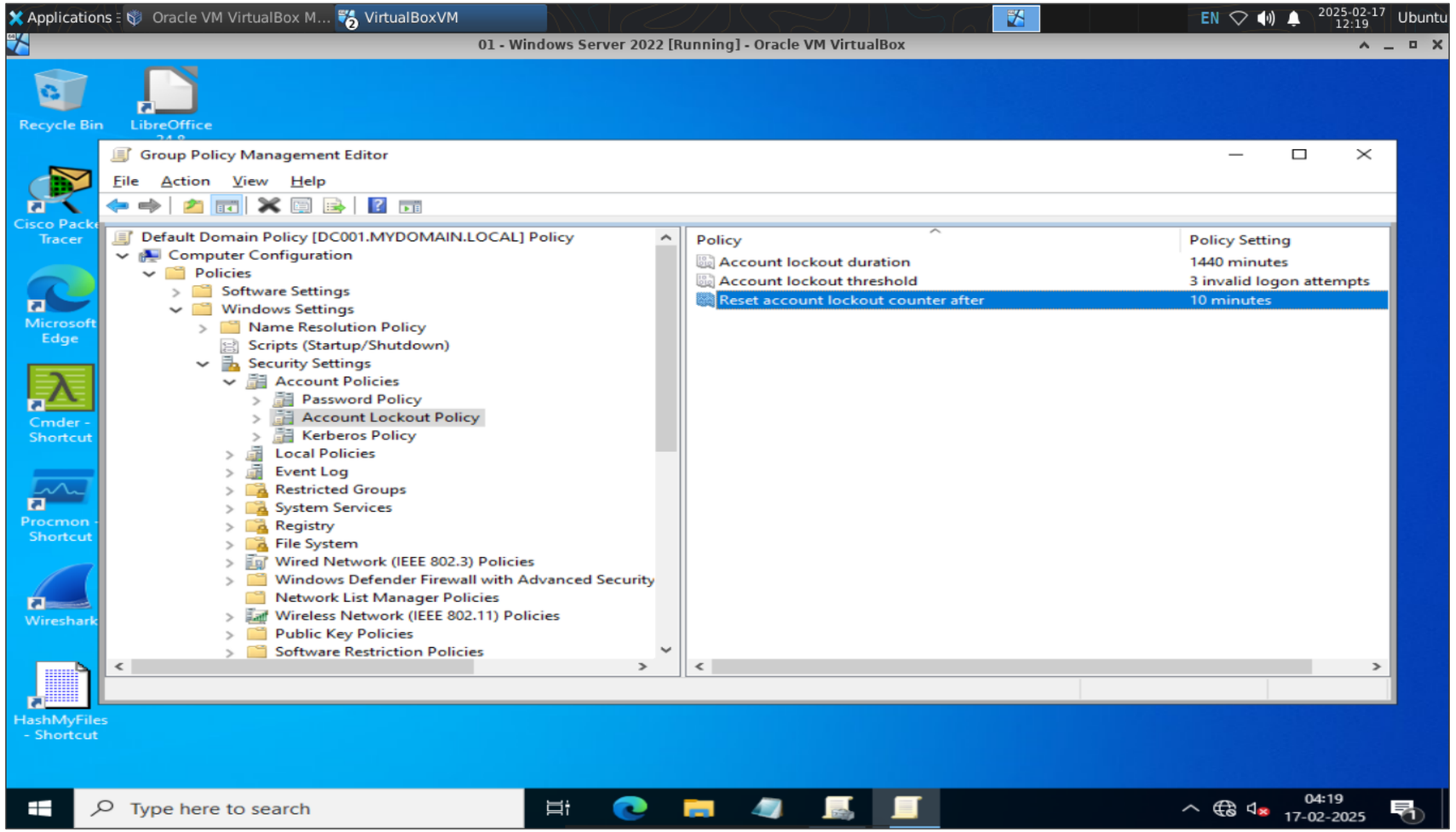
Output Screenshots

The Active Directory configuration was successfully implemented, setting password policies for the Research Organizational Unit (OU) using Group Policy Management Editor. Below is the output:



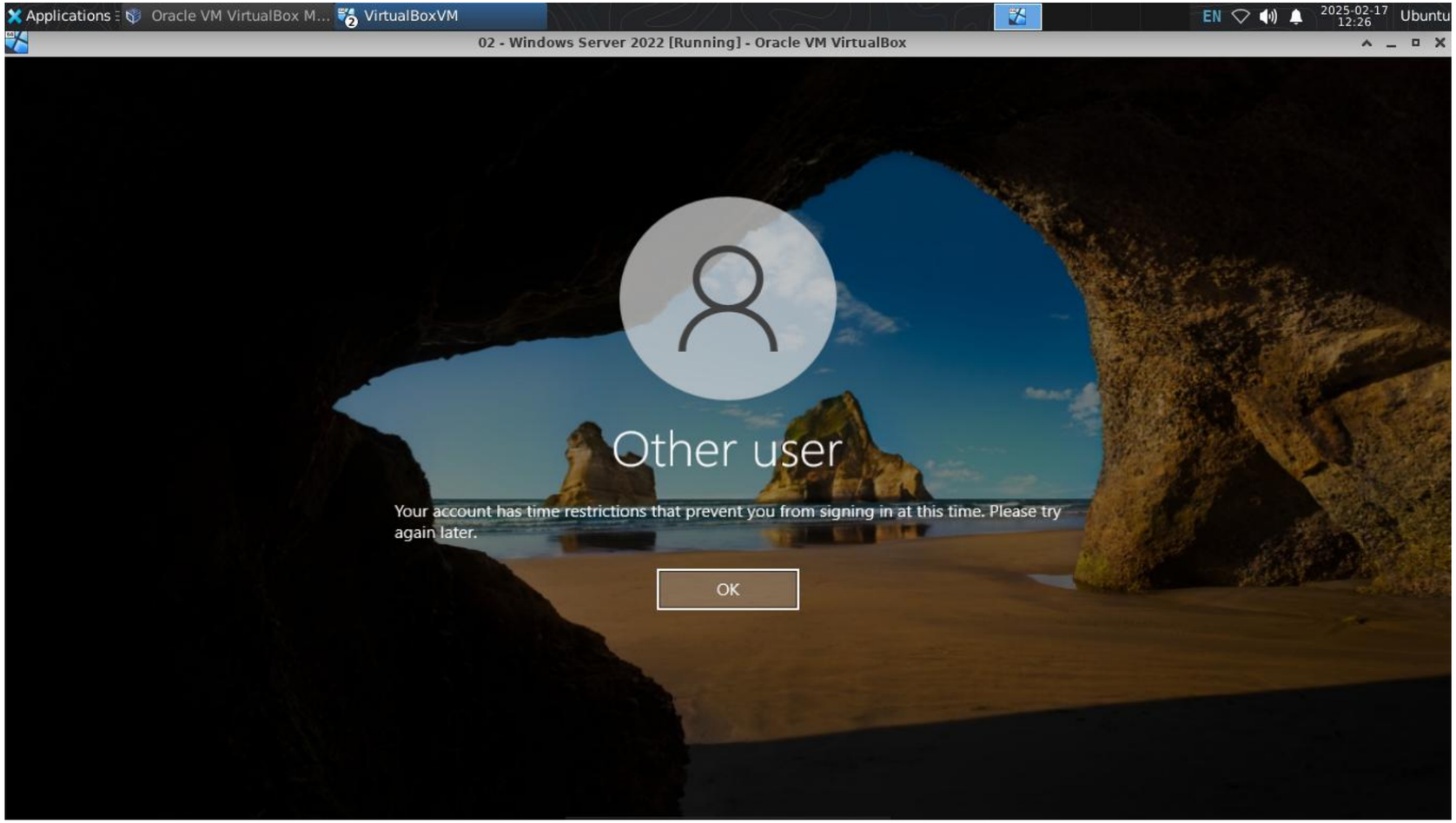
Output Screenshots

The Group Policy Management Editor was configured to enforce Account Lockout Policies under the Default Domain Policy, managing invalid logon attempts for enhanced security.



Output Screenshots

The Group Policy login restrictions were successfully applied, preventing user access outside the allowed time frame. The screenshot shows a login attempt blocked due to time restrictions:



Project References



- Task 1: Enterprise Infrastructure Security
- Task 2: Enterprise Infrastructure Security
- Task 3: Enterprise Infrastructure Security
- Task 4: Enterprise Infrastructure Security
- Task 5: Enterprise Infrastructure Security

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Thank you